

CLab 9

Stephanie Salgado

Setup



/ Labs / CLab9_Stephanie_Salgado /	
Name	Last Modified
 CLab9.ipynb	10 seconds ago

I begin by adding the provided “CLab9.ipynb” file to my JupyterHub.

Reserving resources

```
from fabrictestbed_extensions.fablib.fablib import FablibManager as fablib_manager
fablib = fablib_manager()
conf = fablib.show_config()
```

FABlib Config	
Orchestrator	orchestrator.fabric-testbed.net
Credential Manager	cm.fabric-testbed.net
Core API	uis.fabric-testbed.net
Token File	/home/fabric/tokens.json
Project ID	a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
Bastion Host	bastion.fabric-testbed.net
Bastion Username	ssalg03_0000228084
Bastion Private Key File	/home/fabric/work/fabric_config/fabric-bastion-key
Slice Public Key File	/home/fabric/work/fabric_config/slice_key.pub
Slice Private Key File	/home/fabric/work/fabric_config/slice_key
Sites to avoid	
SSH Command Line	ssh -i {{{_self_private_ssh_key_file}}} -F /home/fabric/work/fabric_config/ssh_config {{{_self_username}}}@{{{_self_management_ip}}}
Log Level	INFO
Log File	/tmp/fablib/fablib.log
Bastion SSH Config File	/home/fabric/work/fabric_config/ssh_config
Version	1.7.3
Data directory	/tmp/fablib

I run the cell to reserve resources for my slice.

```
import os
slice_name="Stephanie_Salgado"+"secure-applications_" + os.getenv('NB_USER')

try:
    slice = fablib.get_slice(slice_name)
    print("You already have a slice by this name!")
    print("If you previously reserved resources, skip to the 'log in to resources' section.")
except:
    print("You don't have a slice named %s yet." % slice_name)
    print("Continue to the next step to make one.")
    slice = fablib.new_slice(name=slice_name)
```

You don't have a slice named Stephanie_Salgadosecure-applications_ssalg03 yet.
Continue to the next step to make one.

Then, I set a name for the slice and check if I already have one named that.

Site	
Name	EDUKY
State	Active
Address	301 Hilltop Avenue, Lexington, KY 40506
Location	(38.0325, -84.502801)
PTP Capable	True
Hosts	18
CPUs	36
Cores Available	73241
Cores Capacity	73728
Cores Allocated	487
Ram Available	7186

Next, I print details about the site.

```
# this cell sets up the host, router and server
node_names = ["romeo", "router", "server"]
for n in node_names:
    slice.add_node(name=n, site=site_name, cores=2, ram=4, disk=10, image='default_ubuntu_20')

# this cell sets up the network links
nets = [
    {"name": "net0", "nodes": ["romeo", "router"]},
    {"name": "net1", "nodes": ["router", "server"]}
]
for n in nets:
    ifaces = [slice.get_node(node).add_component(model="NIC_Basic", name=n["name"]).get_interfaces()[0] for node in n['nodes']]
    slice.add_l2network(name=n["name"], type='L2Bridge', interfaces=ifaces)
```

Then, I run these 2 cells to set up the nodes and network links.

```
slice.submit()
```

Retry: 10, Time: 253 sec

Slice	
ID	6c84daa6-1b79-48e2-9b60-8357605b1d43
Name	Stephanie_Salgadosecure-applications_ssalg03
Lease Expiration (UTC)	2024-11-09 16:42:48 +0000
Lease Start (UTC)	2024-11-08 16:42:48 +0000
Project ID	a70de2f5-9e12-4b6b-b412-0ae1a2c553b0
State	StableOK

```
slice.wait_ssh(progress=True)
```

```
Waiting for slice . Slice state: StableOK
Waiting for ssh in slice . ssh successful
True
```

Then, I submitted the slice.

Configuring resources

```
from ipaddress import ip_address, IPv4Address, IPv4Network

if_conf = {
    "romeo-net0-p1": {"addr": "10.10.1.100", "subnet": "10.10.1.0/24", "hostname": "romeo"},
    "router-net0-p1": {"addr": "10.10.1.1", "subnet": "10.10.1.0/24", "hostname": "router"},
    "router-net1-p1": {"addr": "10.10.2.1", "subnet": "10.10.2.0/24", "hostname": "router"},
    "server-net1-p1": {"addr": "10.10.2.100", "subnet": "10.10.2.0/24", "hostname": "server"}
}

for iface in slice.get_interfaces():
    if_name = iface.get_name()
    hostname = if_conf[if_name]['hostname']
    iface.ip_addr_add(addr=if_conf[if_name]['addr'], subnet=IPv4Network(if_conf[if_name]['subnet']))

slice.get_node(name='romeo').execute(f"echo '{if_conf['server-net1-p1']['addr']}\t{if_conf['server-net1-p1']['hostname']}' | sudo tee -a /etc/hosts > /dev/null")
slice.get_node(name='server').execute(f"echo '{if_conf['romeo-net0-p1']['addr']}\t{if_conf['romeo-net0-p1']['hostname']}' | sudo tee -a /etc/hosts > /dev/null")
slice.get_node(name='router').execute(f"echo '{if_conf['romeo-net0-p1']['addr']}\t{if_conf['romeo-net0-p1']['hostname']}' | sudo tee -a /etc/hosts > /dev/null")
slice.get_node(name='router').execute(f"echo '{if_conf['server-net1-p1']['addr']}\t{if_conf['server-net1-p1']['hostname']}' | sudo tee -a /etc/hosts > /dev/null")
```

```
for iface in slice.get_interfaces():
    iface.ip_link_up()
```

I configure the IP addresses and bring the interfaces up.

```
rt_conf = [
    {"name": "romeo", "addr": "10.10.2.0/24", "gw": "10.10.1.1"},
    {"name": "server", "addr": "10.10.1.0/24", "gw": "10.10.2.1"},
]

for rt in rt_conf:
    slice.get_node(name=rt['name']).ip_route_add(subnet=IPv4Network(rt['addr']), gateway=rt['gw'])
```

```
for n in ['router']:
    slice.get_node(name=n).execute("sudo sysctl -w net.ipv4.ip_forward=1")

net.ipv4.ip_forward = 1
```

After that, I add the routes and enable IP forwarding on the router.

Logging into resources

```
$ ubuntu@romeo: ~
$ ubuntu@router: ~
$ ubuntu@server: ~

ubuntu@romeo:~$
```

Finally, I use the SSH commands to log into each node.

Remote login

Installing the telnet service on "server"

```
ubuntu@server:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:2 http://nova.clouds.archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3289 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [483 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.3 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3227 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [451 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [548 B]
```

```
ubuntu@server:~$ sudo apt -y install xinetd telnetd
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  update-inetd
The following NEW packages will be installed:
  telnetd update-inetd xinetd
```

First, I update then I install

Creating the telnet configuration file

```
ubuntu@server:~$ sudo nano /etc/xinetd.d/telnet
ubuntu@server:~$ cat /etc/xinetd.d/telnet
# default: on
# description: telnet server
service telnet
{
  disable = no
  flags = REUSE
  socket_type = stream
  wait = no
  user = root
  only_from = 10.0.0.0/8
  server = /usr/sbin/in.telnetd
  log_on_failure += USERID
}
```

Restarting the service and checking status

```
ubuntu@server:~$ sudo service xinetd restart
ubuntu@server:~$ service xinetd status
● xinetd.service - LSB: Starts or stops the xinetd daemon.
   Loaded: loaded (/etc/init.d/xinetd; generated)
   Active: active (running) since Fri 2024-11-08 17:20:25 UTC; 7s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 35166 ExecStart=/etc/init.d/xinetd start (code=exited, status=0/SUCCESS)
    Tasks: 1 (limit: 4663)
   Memory: 832.0K
    CGroup: /system.slice/xinetd.service
            └─35192 /usr/sbin/xinetd -pidfile /run/xinetd.pid -stayalive -inetd_compat -inetd_ipv6

Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/discard-udp [file=/etc/xinetd.d/discard-udp] [line=25]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/echo [file=/etc/xinetd.d/echo] [line=14]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/echo-udp [file=/etc/xinetd.d/echo-udp] [line=26]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/servers [file=/etc/xinetd.d/servers] [line=14]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/services [file=/etc/xinetd.d/services] [line=13]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/telnet [file=/etc/xinetd.d/telnet] [line=13]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/time [file=/etc/xinetd.d/time] [line=13]
Nov 08 17:20:25 server xinetd[35192]: Reading included configuration file: /etc/xinetd.d/time-udp [file=/etc/xinetd.d/time-udp] [line=28]
Nov 08 17:20:25 server xinetd[35192]: 2.3.15.3 started with libwrap loadavg labeled-networking options compiled in.
Nov 08 17:20:25 server xinetd[35192]: Started working: 1 available service
```

Notice the “active (running)” message.

Starting an SSH server process on “server”

```
ubuntu@server:~$ sudo /usr/sbin/sshd -o ListenAddress=10.10.2.100 -o PasswordAuthentication=yes -f /usr/share/openssh/sshd_config -p 1000
```

Setting up a user account for remote access

```
ubuntu@server:~$ sudo useradd -m shakespeare -s /bin/sh
ubuntu@server:~$ sudo passwd shakespeare
New password:
Retype new password:
passwd: password updated successfully
```

I added an account with user “shakespeare” and set a password for it.

Capturing traffic (telnet)

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-telnet-$(hostname -s).pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

On “romeo”, I use the command above to capture traffic on the network segment. This traffic will be saved to a file called “security-telnet-...”.

Initiating a telnet connection from "romeo" to "server"

```
ubuntu@romeo:~$ telnet server
Trying 10.10.2.100...
Connected to server.
Escape character is '^'.
Ubuntu 20.04.6 LTS
server login: shakespeare
Password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-186-generic x86_64)
```

I start a connection on "romeo" and log in using the credentials I set up.

```
$ date
Fri Nov  8 17:46:59 UTC 2024
$ exit
Connection closed by foreign host.
```

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-telnet-${hostname -s}.pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
^C135 packets captured
135 packets received by filter
0 packets dropped by kernel
```

Once logged in, I get the date and then exit. I go back and stop the capture.

Capturing traffic (ssh)

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-ssh-${hostname -s}.pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

On "romeo", I capture the traffic on the network segment once more, but this time, the output will be saved in a file called "security-ssh-...".

Initiating an SSH connection

```
ubuntu@romeo:~$ ssh shakespeare@server -p 1000
The authenticity of host '[server]:1000 ([10.10.2.100]:1000)' can't be established.
ECDSA key fingerprint is SHA256:U5XmqjXRmRIImnUxvXoReFoLoKuDNc/cespR4KIUtYjk.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server]:1000,[10.10.2.100]:1000' (ECDSA) to the list of known hosts.
shakespeare@server's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.4.0-186-generic x86_64)
```

I initiate an SSH connection from "romeo" to "server" on port 1000 and log in.

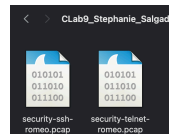
```
$ date
Fri Nov  8 18:16:03 UTC 2024
$ exit
Connection to server closed.
```

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-ssh-$(hostname -s).pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
^C100 packets captured
100 packets received by filter
0 packets dropped by kernel
```

I got the date and exited like before, then stopped the capture.

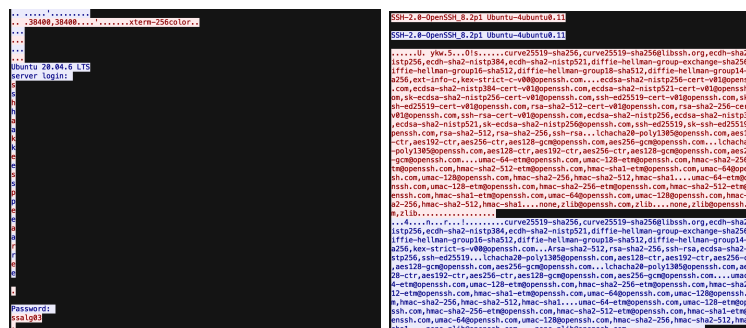
Downloading the capture files

/	
Name	Last Modified
fabric_config	22 days ago
jupyter-examples-rel1.7.0	last month
Labs	2 hours ago
security-ssh-romeo.pcap	2 minutes ago
security-telnet-romeo.pcap	2 minutes ago



I run the cells to transfer the “.pcap” files and download them to my Mac.

Using TCP Stream on one of the packets from both files



I opened the files, then selected a TCP packet (packet 10). From there, I went to “Analyze” > “Follow” > “TCP Stream”. There was a big difference. The telnet (left) showed the password used. While the ssh (right) was really hard to read.

File transfer

Installing SFTP server on "server"

```
ubuntu@server:~$ sudo apt -y install vsftpd
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ssl-cert
Suggested packages:
  openssl-blacklist
The following NEW packages will be installed:
  ssl-cert vsftpd
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 132 kB of archives.
```

Capturing traffic (ftp)

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-ftp-$(hostname -s).pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

I start capturing traffic on "romeo".

Initiating an FTP session

```
ubuntu@romeo:~$ ftp server
Connected to server.
220 (vsFTPd 3.0.5)
Name (server:ubuntu): shakespeare
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Notice the "230 Login successful" message.

```
ftp>~cd /etc
250 Directory successfully changed.
ftp> get passwd
local: passwd remote: passwd
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for passwd (1993 bytes).
226 Transfer complete.
1993 bytes received in 0.00 secs (65.5404 MB/s)
ftp> exit
221 Goodbye.
```

I go into "/etc", use "get passwd", and exit after that. Lastly, I stop the capture.


```
ubuntu@romeo:~$ ls
passwd  security-ftp-romeo.pcap  security-ssh-romeo.pcap  security-telnet-romeo.pcap
```

```
ubuntu@romeo:~$ cat passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106:,:/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/:home/syslog:/usr/sbin/nologin
_apt:x:105:65534:/:nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uuid:x:107:112:/:run/uuid:/usr/sbin/nologin
tcpdump:x:108:113:/:nonexistent:/usr/sbin/nologin
sshd:x:109:65534:/:run/ssh:/usr/sbin/nologin
landscape:x:110:115:/:var/lib/landscape:/usr/sbin/nologin
pollinate:x:111:1:/:var/cache/pollinate:/bin/false
fwupd-refresh:x:112:116:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
lxd:x:998:100:/:var/snap/lxd/common/lxd:/bin/false
telnetd:x:113:120:/:nonexistent:/usr/sbin/nologin
shakespeare:x:1001:1001:/:home/shakespeare:/bin/sh
ftp:x:114:122:ftp daemon,,,:/srv/ftp:/usr/sbin/nologin
```

I use “ls” and aside from the newly added packet capture file, the “passwd” file I transferred is there. I use “cat” to view its contents.

Capturing traffic (sftp)

```
ubuntu@romeo:~$ sudo tcpdump -i $(ip route get 10.10.2.100 | grep -oP "(?<=dev )[^ ]+") -w security-sftp-$(hostname -s).pcap
tcpdump: listening on enp7s0, link-type EN10MB (Ethernet), capture size 262144 bytes
```

Once again, I begin capturing traffic on the network segment.

Initiating an SFTP connection

```
ubuntu@romeo:~$ sftp -P 1000 shakespeare@server
shakespeare@server's password:
Connected to server.
sftp>
```

I initiate an SFTP connection and login with the appropriate credentials.

```
sftp> cd /etc
sftp> get passwd
Fetching /etc/passwd to passwd
/etc/passwd
sftp> exit
```

Then, I navigate to “/etc”, use “get passwd”, and exit. Then I stop the capture.

Downloading the capture files

```
ftp_pcap = "/home/ubuntu/security-ftp-ns.pcap" % slice.get_node("romeo").execute("hostname", quiet=True)[0].strip()
slice.get_node("romeo").download_file("/home/fabric/work/security-ftp-romeo.pcap", ftp_pcap)

sftp_pcap = "/home/ubuntu/security-sftp-ns.pcap" % slice.get_node("romeo").execute("hostname", quiet=True)[0].strip()
slice.get_node("romeo").download_file("/home/fabric/work/security-sftp-romeo.pcap", sftp_pcap)
```

Name	Last Modified
fabric_config	22 days ago
jupyter-examples-rel1.7.0	last month
Labs	3 hours ago
security-ftp-romeo.pcap	40 seconds ago
security-sftp-romeo.pcap	37 seconds ago
security-ssh-romeo.pcap	13 minutes ago
security-telnet-romeo.pcap	13 minutes ago

I run the appropriate cells and download both capture files.

Using TCP Stream on one of the packets from both files

```
220 (vsFTPD 3.0.5)
USER shakespeare
331 Please specify the password.
PASS ssalg03
230 Login successful.
SYST
215 UNIX Type: L8
CWD /etc
250 Directory successfully changed.
TYPE I
200 Switching to Binary mode.
PORT 10,10,1,100,142,71
200 PORT command successful. Consider using PASV.
RETR passwd
150 Opening BINARY mode data connection for passwd (1993 bytes).
226 Transfer complete.
QUIT
221 Goodbye.
```

```
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
.....U..A....1...i*....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-n
istp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,d
iffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sh
a256,ext-info-c,kex-strict-c-v00@openssh.com,...ecdsa-sha2-nistp256-cert-v01@openssh
.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.c
om,sk-ecdsa-sha2-nistp256-cert-v01@openssh.com,ssh-ed25519-cert-v01@openssh.com,sk-s
sh-ed25519-cert-v01@openssh.com,rsa-sha2-512-cert-v01@openssh.com,rsa-sha2-256-cert-
v01@openssh.com,ssh-rsa-cert-v01@openssh.com,ecdsa-sha2-nistp256,ecdsa-sha2-nistp384
,ecdsa-sha2-nistp521,sk-ecdsa-sha2-nistp256@openssh.com,ssh-ed25519,sk-ssh-ed25519@o
penssh.com,rsa-sha2-512,rsa-sha2-256,ssh-rsa...lchacha20-poly1305@openssh.com,aes128
-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com...lchacha20
-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256
-gcm@openssh.com...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-e
tm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@ope
nssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha1...umac-64-etm@ope
nssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@op
enssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sh
a2-256,hmac-sha2-512,hmac-sha1...none,zlib@openssh.com,zlib...none,zlib@openssh.co
m,zlib.....
...4..Z'.n.(..`A)m.?.....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-n
istp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,d
iffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sh
a256,kex-strict-s-v00@openssh.com...Arsa-sha2-512,rsa-sha2-256,ssh-rsa,ecdsa-sha2-ni
stp256,ssh-ed25519...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr
,aes128-gcm@openssh.com,aes256-gcm@openssh.com...lchacha20-poly1305@openssh.com,aes1
28-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com...umac-6
4-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-5
12-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.c
om,hmac-sha2-256,hmac-sha2-512,hmac-sha1...umac-64-etm@openssh.com,umac-128-etm@ope
nssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@op
enssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-
sha1...none,zlib@openssh.com...none,zlib@openssh.com.....
```

I selected a packet (packet 70) on both capture files. Then went to “Analyze” > “Follow” > “TCP Stream”. I noticed that ftp (left) also displayed the user and password very easily, while sftp (right) looked a lot like the ssh capture.

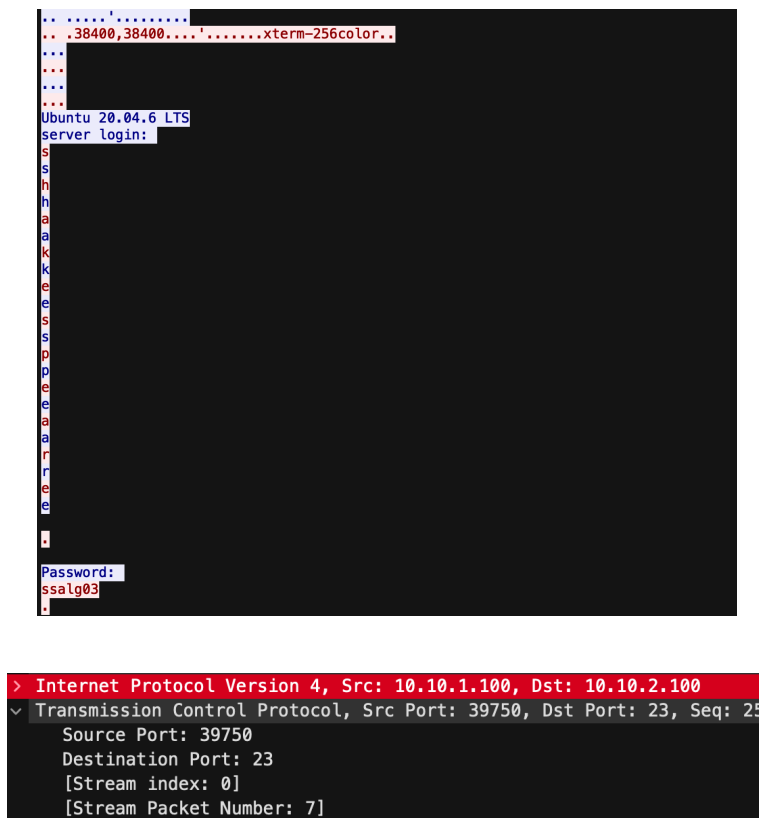
Deleting the slice

Stephanie_Salgadosecure-applications_ssalg03	Dead	2024-11-09 10:42:48	Computer_Networks	 Slice ID
--	------	------------------------	-------------------	--

Lastly, I run the cell to delete the slice.

Exercises

1. In the packet capture of the telnet experiment, can you read: the username and password? IP/TCP headers? Session data? Show evidence.



The image shows two screenshots. The top screenshot is a terminal window showing a telnet session on Ubuntu 20.04.6 LTS. The prompt is 'server login:' and the user has entered the username 'ssalg03'. The prompt 'Password:' is visible, and the user has entered a password. The bottom screenshot is a packet capture analysis showing the IP and TCP headers for a packet. The IP header shows Source: 10.10.1.100 and Destination: 10.10.2.100. The TCP header shows Source Port: 39750 and Destination Port: 23. The stream index is 0 and the stream packet number is 7.

```
> Internet Protocol Version 4, Src: 10.10.1.100, Dst: 10.10.2.100
Transmission Control Protocol, Src Port: 39750, Dst Port: 23, Seq: 25
  Source Port: 39750
  Destination Port: 23
  [Stream index: 0]
  [Stream Packet Number: 7]
```

As included above, the telnet capture file actually stored the username and password in plaintext, making them easily readable. The IP/TCP headers and session data were also visible.

2. In the packet capture of the SSH experiment, can you read: the username and password? IP/TCP headers? Session data? Show evidence.

```
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.11
.....U. ykw.5...0!s.....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-n
istp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,d
iffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sh
a256,ext-info-c,kex-strict-c-v00@openssh.com....ecdsa-sha2-nistp256-cert-v01@openssh
.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.c
om,sk-ecdsa-sha2-nistp256-cert-v01@openssh.com,ssh-ed25519-cert-v01@openssh.com,sk-s
sh-ed25519-cert-v01@openssh.com,rsa-sha2-512-cert-v01@openssh.com,rsa-sha2-256-cert-
v01@openssh.com,ssh-rsa-cert-v01@openssh.com,ecdsa-sha2-nistp256,ecdsa-sha2-nistp384
,ecdsa-sha2-nistp521,sk-ecdsa-sha2-nistp256@openssh.com,ssh-ed25519,sk-ssh-ed25519@o
penssh.com,rsa-sha2-512,rsa-sha2-256,ssh-rsa...lchacha20-poly1305@openssh.com,aes128
-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com...lchacha20
-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256
-gcm@openssh.com...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-e
tm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@ope
nssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1...umac-64-etm@ope
nssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@op
enssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sh
a2-256,hmac-sha2-512,hmac-sha1...none,zlib@openssh.com,zlib...none,zlib@openssh.co
m,zlib.....
...4....n...r...!.....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-n
istp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,d
iffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sh
a256,kex-strict-s-v00@openssh.com...Arsa-sha2-512,rsa-sha2-256,ssh-rsa,ecdsa-sha2-ni
stp256,ssh-ed25519...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr
,aes128-gcm@openssh.com,aes256-gcm@openssh.com...lchacha20-poly1305@openssh.com,aes1
28-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com...umac-6
4-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-5
12-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.co
m,hmac-sha2-256,hmac-sha2-512,hmac-sha1...umac-64-etm@openssh.com,umac-128-etm@open
ssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@op
enssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-
sha1...none,zlib@openssh.com...none,zlib@openssh.com.....
```

```
> Internet Protocol Version 4, Src: 10.10.1.100, Dst: 10.10.2.100
v Transmission Control Protocol, Src Port: 51920, Dst Port: 1000, Seq: 43,
  Source Port: 51920
  Destination Port: 1000
  [Stream index: 0]
  [Stream Packet Number: 7]
```

Here, the username and password aren't visible because data is encrypted. However, you can still view IP/TCP headers and session data.

3. In the packet capture of the FTP experiment, can you read: the username and password? IP/TCP headers? The name of the file transferred, and the file contents? Show evidence.

```
220 (vsFTPd 3.0.5)
USER shakespear
331 Please specify the password.
PASS ssa1g03
230 Login successful.
SYST
215 UNIX Type: L8
CWD /etc
250 Directory successfully changed.
TYPE I
200 Switching to Binary mode.
PORT 30,30,3,100,142,71
200 PORT command successful. Consider using PASV.
RETR passwd
150 Opening BINARY mode data connection for passwd (1993 bytes).
226 Transfer complete.
QUIT
221 Goodbye.
```

With FTP, the username and password were visible. The name of the file transferred is also seen, however I wasn't able to find where the file contents might have been visible.

4. In the packet capture of the SFTP experiment, can you read: the username and password? IP/TCP headers? The name of the file transferred, and the file contents? Show evidence.

```
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.1
SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.1
.....U.....K.....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,ext-info-c,kex-strict-c-v00@openssh.com,...ecdsa-sha2-nistp256-cert-v01@openssh.com,ecdsa-sha2-nistp256-cert-v01@openssh.com,ecdsa-sha2-nistp384-cert-v01@openssh.com,ecdsa-sha2-nistp521-cert-v01@openssh.com,sk-ecdsa-sha2-nistp256-cert-v01@openssh.com,ssh-ed25519-cert-v01@openssh.com,sk-sha256-cert-v01@openssh.com,rsa-sha2-512-cert-v01@openssh.com,rsa-sha2-256-cert-v01@openssh.com,ssh-rsa-cert-v01@openssh.com,ecdsa-sha2-nistp384,ecdsa-sha2-nistp521,sk-ecdsa-sha2-nistp256@openssh.com,ssh-ed25519,sk-ssh-ed25519@openssh.com,rsa-sha2-512,rsa-sha2-256,ssh-rsa,...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1,...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1,...none,zlib@openssh.com,zlib,...none,zlib@openssh.com,zlib.....
.....K.....curve25519-sha256,curve25519-sha256@libssh.org,ecdh-sha2-nistp256,ecdh-sha2-nistp384,ecdh-sha2-nistp521,diffie-hellman-group-exchange-sha256,diffie-hellman-group16-sha512,diffie-hellman-group18-sha512,diffie-hellman-group14-sha256,kex-strict-c-v00@openssh.com,...Arsha-sha2-512,rsa-sha2-256,ssh-rsa,ecdsa-sha2-nistp256,ssh-ed25519,...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,...lchacha20-poly1305@openssh.com,aes128-ctr,aes192-ctr,aes256-ctr,aes128-gcm@openssh.com,aes256-gcm@openssh.com,...umac-64-etm@openssh.com,umac-128-etm@openssh.com,hmac-sha2-256-etm@openssh.com,hmac-sha2-512-etm@openssh.com,hmac-sha1-etm@openssh.com,umac-64@openssh.com,umac-128@openssh.com,hmac-sha2-256,hmac-sha2-512,hmac-sha1,...none,zlib@openssh.com,zlib.....
```

With the SFTP experiment, the username and password were once again not readable. I was unable to find the name of the file transferred or the contents since that information isn't readily available. I also noticed the packets of this capture file were all TCP unlike the FTP experiment.

Summary

In this lab, I captured and analyzed network traffic for Telnet, SSH, FTP, and SFTP sessions to observe differences in data security. For Telnet, both the username and password were visible in plaintext, along with session data. SSH, however, encrypted session details, making credentials unreadable while still showing IP/TCP headers. During the FTP session, I observed that the username, password, and file name were transmitted in plaintext. In contrast, SFTP encrypted all sensitive information, securing both login credentials and file details. This lab demonstrated how encryption in SSH and SFTP enhances security by protecting data, unlike Telnet and FTP, which leave it exposed. Overall, I think this lab was quite simple.